RISK ASSESSMENT

PSEUDOCODE

```
BEGIN
DEFINE FUNCTION risk_score
INPUT Likelihood in a scale (1-5)
INPUT Consequence in a scale (1-5)
Calculate Risk_Score:
        Risk_Score = Likelihood*Consequence
IF Risk_Score < 0 or Risk_Score >25:
        Classification = “Invalid”
        PRINT(“Invalid Risk Score”)
ELIF Risk_Score <= 4:
        Classification = “Low”
ELIF Risk_Score <=9:
        Classification = “Medium”
ELIF Risk_Score <=16:
        Classification = “High”
ELSE:
        Classification = “Critical”
PRINT(“Risk Score:”, Risk_Score, “Classification:”, Classification)
IF Classification == “High”:
        PRINT(“WARNING! High Risk identified; immediate attention required”)
ELIF Classification== “Critical”:
        PRINT(“WARNING! Critical Risk identified; immediate attention required”)
END FUNCTION
CALL FUNCTION risk_score
END
```

ALERTS AND NOTIFICATIONS

PSEUDOCODE

```
BEGIN
DEFINE FUNCTION monitoring_alerts
INPUT Equipment ID
IF Equipment ID does not exist THEN
   DISPLAY "Equipment ID not found"
ELSE
   DISPLAY Equipment ID and Equipment Type
   IF Temperature > 100 THEN
      DISPLAY "CRITICAL: High temperature"
```

```
    ELSE IF Temperature >= 80 THEN
        DISPLAY "WARNING: High temperature"
    END IF
    IF Vibration > 8 THEN
        DISPLAY "CRITICAL: High vibration"
    ELSE IF Vibration >= 5 THEN
        DISPLAY "WARNING: High vibration"
    END IF
    IF Downtime > 24 THEN
        DISPLAY "WARNING: Excessive equipment downtime"
    END IF
    IF Maintenance Status = "Due" OR "Overdue" THEN
        DISPLAY "WARNING: Maintenance due or overdue"
    END IF

    IF Critical Safety Incidents are present THEN
        DISPLAY "NOTICE: Critical safety incidents recorded"
    END IF
    IF High-Risk Safety Observations are observed THEN
        DISPLAY "NOTICE: High-risk safety observations recorded"
    END IF
    IF High or Critical Risk Assessments are present THEN
        DISPLAY "NOTICE: High or critical risks recorded"
    END IF
    IF No alerts were generated THEN
        DISPLAY "No alerts, all conditions normal"
    END IF
END IF
CALL FUNCTION monitoring_alerts
END
```